AUTO GAUGE / REVISION A0 / SCHEMATIC FOR ENGINEERING REVIEW

Nominal 12 V OBD CAN only — one custom PCB — 1.75 inch AMOLED touch

User authorized schematic progression (DEC-009). K0 is NOT passed. No board layout or manufacture release.

OPEN: panel FPC/rails, automotive TVS/FET/SOA/limits, wake sequencing and IGN discrimination, USB protection/current.

TBD values and OPEN pins are deliberate blockers. ERC exceptions are reported, not hidden.

01_OBD_INPUT

File: 01_OBD_INPUT.kicad_sch

02_AUTOMOTIVE_POWER

File: 02_AUTOMOTIVE_POWER.kicad_sch

03_MCU_MEMORY_BLE

File: 03_MCU_MEMORY_BLE.kicad_sch

04_CAN_PHY

File: 04_CAN_PHY.kicad_sch

05_AMOLED_TOUCH

File: 05_AMOLED_TOUCH.kicad_sch

06_LIGHT_SENSOR

File: 06_LIGHT_SENSOR.kicad_sch

07_TEST_SERVICE

File: 07_TEST_SERVICE.kicad_sch

Sheet: /
File: auto_gauge.kicad_sch

Title:

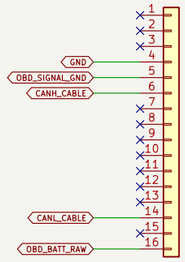
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 1/8

REV A0 / ENGINEERING REVIEW — NOT FOR BUILD

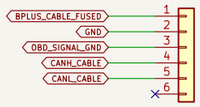
Vehicle plug and 2 m harness are external wiring, not a second PCB.

J101 pin assignment is a proposed gauge-side map; mating orientation and actual MPN remain OPEN.

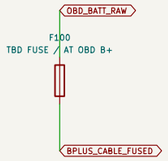
J100
EXTERNAL J1962 TYPE A / 12V

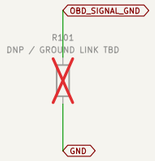


J101
GAUGE INPUT / MAP H



#FLG01
PWR_FLAG





#FLG02
PWR_FLAG





No OEM pins. No K-line/J1850. CAN_H/L twisted pair; NO 120 ohm termination in this gauge.

F100 physically close to OBD pin 16. Fuse value, cable gauge and strain relief require fault/inrush analysis.

OBD pin 4 is the proposed power return. Pin 5 remains separate; R101 is DNP pending ground/EMC review.

Do not treat the illustrative 6-way pin numbering as a verified manufacturer mating view.

NO VEHICLE CONNECTION OR FABRICATION RELEASE

Sheet: /01_OBD_INPUT/
File: 01_OBD_INPUT.kicad_sch

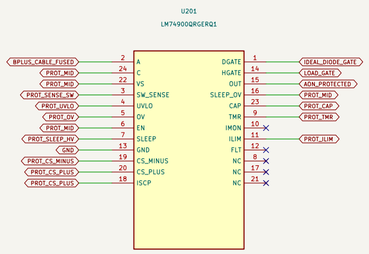
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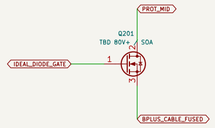
Size: A3	Date: 2026-10-24	**Rev: A0 REVIEW**
KiCad E.D.A. 10.0.7		Id: 2/8

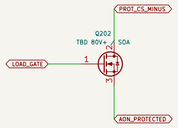
REV A0 / ENGINEERING REVIEW — NOT FOR BUILD

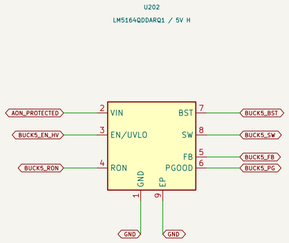
OPEN P0: transient profiles, TVS clamp/energy, MOSFET SOA, limits and wake timing. DO NOT BUILD.

Common-drain topology from E05. Capacitor voltage ratings / passive values with H are engineering assumptions.



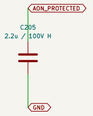






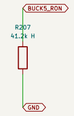


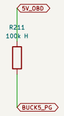


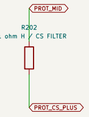


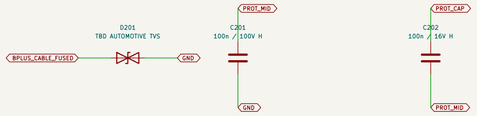


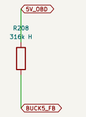


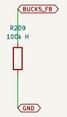


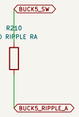


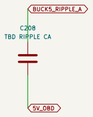


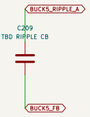




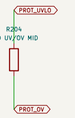


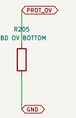


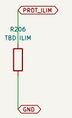






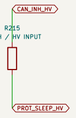


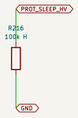












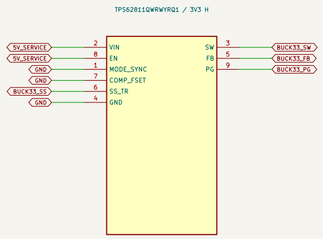


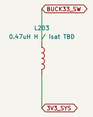


R208/R209 => 4.992V ideal, 4.839..5.145V with ref/divider corners before ripple error.

R207 => about303kHz. R210/C208/C209 stay TBD; see BUCK5_FEEDBACK_REVIEW.md.

12V-tuned ripple network fails 12mV screening near6V. Low-VIN option adds DC error.





3V3 divider => 3.315V. COMP=GND: 2.25MHz.

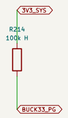
Min Ceff15uF. Validate bias/inrush/loop.



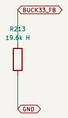














LM74900 EP / RTN must FLOAT. Do not connect its exposed pad to GND (E05).

INH is HV logic, not a source for MCU. Main FET path must be ON before buck inrush; R217/C217 remain TBD.

Sleep bypass is for the CAN always-on load only. It is not a startup supply for MCU/display.

C201 / C202 are per E05. All voltage ratings, surge stress and UV/OV resistor ladder remain under review.

OPEN: sleep OCP latch-off may remove CAN VSUP/INH. EN tied to VS is NOT independent fault recovery.

Complete recovery architecture and test fault return before build: the RC wake delay alone is insufficient.

NO VEHICLE CONNECTION OR FABRICATION RELEASE

Sheet: /02_AUTOMOTIVE_POWER/
File: 02_AUTOMOTIVE_POWER.kicad_sch

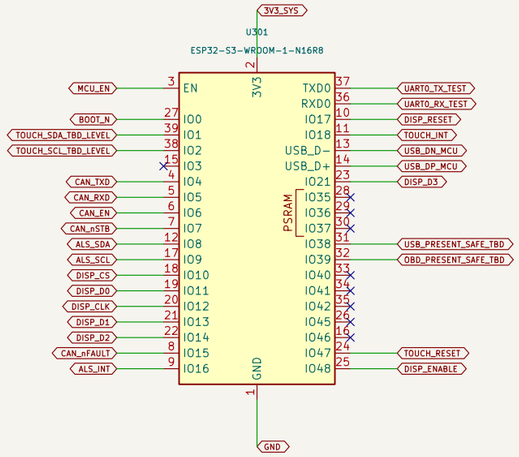
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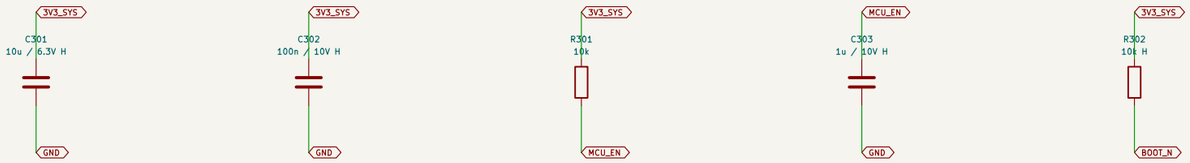
Size: A2 | Date: 2026-10-24 | **Rev: A0 REVIEW**
KiCad E.D.A. 10.0.7 | Id: 3/8

REV A0 / ENGINEERING REVIEW — NOT FOR BUILD

ESP32-S3-WROOM-1-N16R8 candidate. Internal flash/PSRAM: GPIO35/36/37 RESERVED for octal memory.

GPIO assignment H for review. GPIO19/20 reserved for USB; GPIO0/45/46 retain boot roles.







EN RC per E02: 10k/1u initial values. Power-up/brownout timing and signed recovery require bench tests.

NCs on unused/reserved GPIO are deliberate. Do not assign GPIO35/36/37 to the panel on N16R8.

USB_PRESENT_SAFE_TBD / OBD_PRESENT_SAFE_TBD require domain-safe sensing: not raw B+ or raw VBUS.

NO VEHICLE CONNECTION OR FABRICATION RELEASE

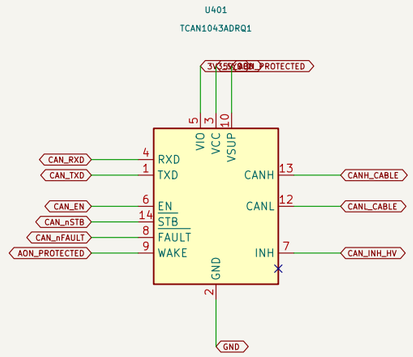
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File: 03_MCU_MEMORY_BLE.kicad_sch

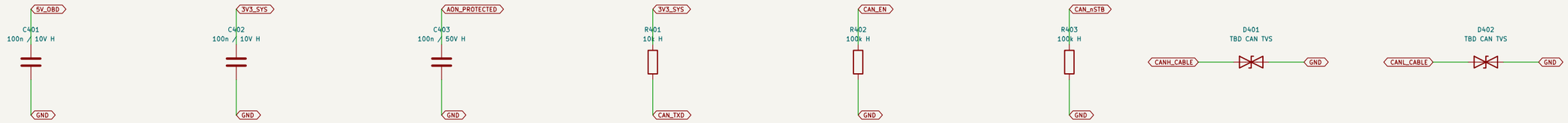
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Size: A3	Date: 2026-10-24	**Rev: A0 REVIEW**
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REV A0 / ENGINEERING REVIEW — NOT FOR BUILD

Candidate TCAN1043ADRQ1; VCC from OBD-only 5V, VIO from MCU 3V3. No added termination.





WAKE tied to VSUP to disable unused local wake input (verify E03). INH goes to HV power control only.

EN/nSTB pull-downs favor non-normal mode on reset. Firmware must remain TX-inhibited until active state qualified.

CAN activity is not IGN. USB-only must not authorize requests; partial-supply leakage and modes require tests.

OPEN: TVS unidirectional/bidirectional network, negative CAN common-mode, capacitance, ESD and 2m signal integrity.

Generic two-terminal TVS symbols are selection placeholders, NOT approval of clamp polarity or a real MPN.

NO VEHICLE CONNECTION OR FABRICATION RELEASE

Sheet: /04_CAN_PHY/
File: 04_CAN_PHY.kicad_sch

Title: Auto Gauge — 04_CAN_PHY

Size: A3	Date: 2026-10-24	**Rev: A0 REVIEW**
KiCad E.D.A. 10.0.7		Id: 5/8

REV A0 / ENGINEERING REVIEW — NOT FOR BUILD

P0 OPEN / DO NOT CONNECT FPC: ESHX0175Y003AMT003 V0 has conflicting rail/temperature data.

J501 is a provisional 31-way logical placeholder, excluded from BOM/board. NO physical pin mapping assigned.



TP501 DISP_CLK → DISP_CLK

TP502 DISP_CS → DISP_CS

TP503 DISP_D0 → DISP_D0

TP504 DISP_D1 → DISP_D1

TP505 DISP_D2 → DISP_D2

TP506 DISP_D3 → DISP_D3

TP507 DISP_RESET → DISP_RESET

TP508 DISP_ENABLE → DISP_ENABLE

TP509 TOUCH_SCL_TBD_LEVEL → TOUCH_SCL_TBD_LEVEL

TP510 TOUCH_SDA_TBD_LEVEL → TOUCH_SDA_TBD_LEVEL

TP511 TOUCH_INT → TOUCH_INT

TP512 TOUCH_RESET → TOUCH_RESET

QSPI and touch levels/rails/sequencing: require corrected supplier specification. No devboard pinout used.

Touch level translation/isolation is OPEN; do not wire the 3V3 GPIO directly to a possibly 1V8 FPC.

Panel rails, panel power switches, FPC pin1/contact orientation and mating connector remain unassigned.

NO VEHICLE CONNECTION OR FABRICATION RELEASE

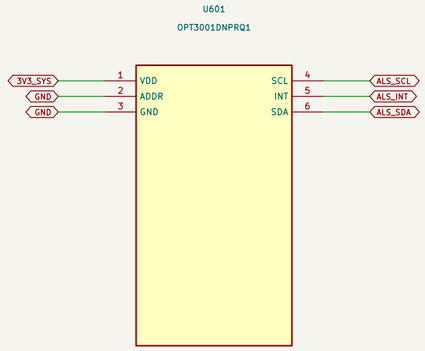
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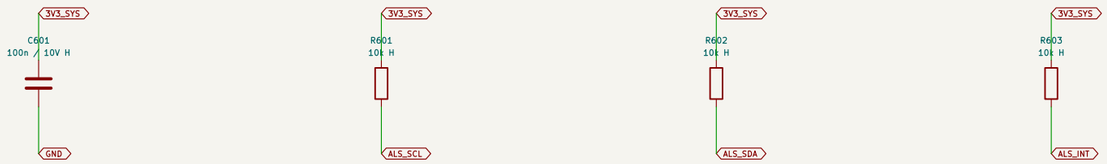
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Size: A3	Date: 2026-10-24	**Rev: A0 REVIEW**
KiCad E.D.A. 10.0.7		Id: 6/8

REV A0 / ENGINEERING REVIEW — NOT FOR BUILD

OPT3001DNPRQ1 / E06 Table4–1. Separate MCU I2C bus avoids assuming panel touch logic levels.





ADDR=GND => address0x44. Validate bus capacitance/rise time before fixing 10k pull-ups.

OPEN: optical window and calibration, thermal pad/land pattern, field of view and light leakage from AMOLED.

Sensor power removed with 3V3_SYS during OBD standby. No always-on ALS wake assumed.

NO VEHICLE CONNECTION OR FABRICATION RELEASE

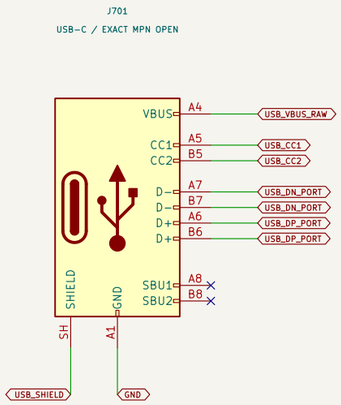
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File: 06_LIGHT_SENSOR.kicad_sch

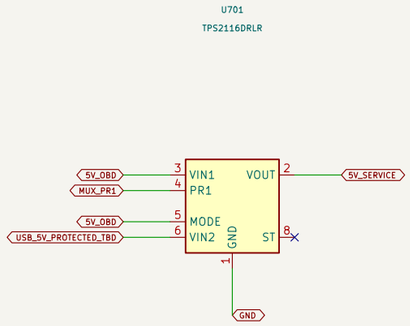
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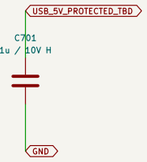
Size: A3	Date: 2026-10-24	**Rev: A0 REVIEW**
KiCad E.D.A. 10.0.7		Id: 7/8

REV A0 / ENGINEERING REVIEW — NOT FOR BUILD

USB-C service only / sink / USB2. Recovery powered without OBD per DEC-007. No PD or source mode.



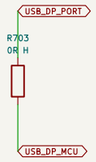


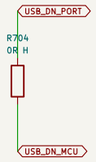




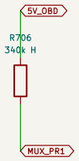


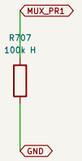












OPEN SERIES FUNCTION: USB_VBUS_RAW -> current limit / OVP / controlled inrush -> USB_5V_PROTECTED_TBD

No wire/jumper bridges that missing protection. CC current detection and ROM-stage power budget remain OPEN.





MUX: OBD priority H at about4.4V nominal (Vref1V: tolerance matters), MODE=VIN1 per E30.

Secure Boot: do not rely on ROM USB-OTG DFU. Validate USB-Serial/JTAG secure downloader + BOOT/RESET.

OPEN: data/CC ESD network and safe USB/OBD presence sensing. No eFuse provisioning or keys generated.

Rd does not grant 1.5A/3A. Verify advertised current and all pre/post-enumeration states including ROM recovery.

NO VEHICLE CONNECTION OR FABRICATION RELEASE

Sheet: /07_TEST_SERVICE/
File: 07_TEST_SERVICE.kicad_sch

Title: Auto Gauge — 07_TEST_SERVICE

Size: A3	Date: 2026-10-24	**Rev: A0 REVIEW**
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